

Rohit Chaudhary

thetuesday96@gmail.com | +91-8988449288 | [Linkedin](#) | [Git](#) | [Portfolio](#)

Backend & AI Engineer with 4+ years of experience building scalable backend systems and Generative AI solutions using Python, FastAPI, MongoDB, LLMs, RAG, and Agentic AI. Experienced in multi-agent architectures using AWS Bedrock and Strands Agents, A2A communication, semantic search, and healthcare AI workflows.

PROFESSIONAL EXPERIENCE AND LEADERSHIP

Persistent Systems Ltd.

Noida, UP

Lead Software Engineer

Feb 2025 – Present

- Designed and developed a multi-agent healthcare solution using Python, AWS Bedrock, and Strands Agents, supporting provider search, provider information, and appointment booking workflows.
- Designed the high-level architecture for scalable Agentic AI workflows, defining agent orchestration, service interactions, A2A communication, session management, and integration with healthcare APIs and data services.
- Built specialized AI agents for provider discovery, provider information, and appointment booking, enabling coordinated multi-step task execution through agent-to-agent (A2A) communication.
- Integrated AWS Bedrock for LLM-powered reasoning and session management, maintaining conversational context across multi-agent interactions and reducing repeated user inputs.
- Developed scalable FastAPI microservices and semantic/vector search capabilities using embeddings and MongoDB to support AI-powered information retrieval and agent workflows.
- Engineered RAG and Agentic AI workflows integrating structured and unstructured data sources, reducing operational effort by up to 70%.
- Optimized MongoDB data models, indexing strategies, and aggregation pipelines, improving query performance by 40% in high-throughput production workloads.

Webner Solutions Pvt. Ltd.

Mohali, Punjab

Software Engineer, Research&Development Division

Jan 2022 – Jan 2025

- Developed scalable backend applications and RESTful APIs using Python, FastAPI, Django, and Flask for production-grade applications.
- Automated API testing workflows using Python-based automation scripts, reducing manual testing effort by 90% and improving release efficiency.
- Designed and optimized REST APIs for data processing, system integration, and backend services with a focus on scalability, performance, and reliability.
- Developed AI-powered chatbot solutions using GPT-3.5/4 and LangChain, implementing natural language querying and NL2SQL capabilities and improving user interaction efficiency by 30%.
- Implemented Generative AI and LLM-based solutions including prompt engineering, conversational AI, and intelligent data retrieval.
- Integrated machine learning and computer vision solutions using OCR, TensorFlow, and OpenCV, improving data extraction and processing efficiency by 40%.
- Collaborated across development, testing, deployment, and system design activities while mentoring junior developers and promoting backend engineering best practices.

Itronix Solutions Pvt. Ltd.

Mohali, Punjab

Python Programming Language and Data Analytics, Internship

- Developed backend APIs and automation scripts using Python, FastAPI, and Flask.
- Supported implementation of machine learning and computer vision features using OCR, TensorFlow, and OpenCV.
- Contributed to debugging, testing, data processing, and application performance improvements.

SKILLS

Languages & Frameworks: Python, JavaScript, SQL

Backend & Frameworks: FastAPI, Django, Flask, Pydantic, RESTful APIs, Microservices, API Design, API Integration

Generative AI & Agentic AI: Generative AI, Agentic AI, AI Agents, LLMs, AWS Bedrock, Strands Agents, Multi-Agent Systems, Agent-to-Agent (A2A) Communication, Tool Calling, Function Calling, RAG, Prompt Engineering, LLM Integration, Conversational AI

AI & Search: Semantic Search, Vector Search, Lexical Search, Hybrid Search, Vector Embeddings, Information Retrieval, Natural Language Processing, NL2SQL

Databases: MongoDB, MongoDB Atlas, PostgreSQL, MySQL, SQLite, Database Indexing, Aggregation Pipelines, Data Modeling

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Git, GitLab, Postman

Databases: MongoDB (Indexing, Aggregation Pipelines, Atlas), PostgreSQL, MySQL, SQLite

System Design: Microservices, Distributed Systems, REST Architecture, API Design, Scalable Backend Systems

EDUCATION

Indus International University, School of Computer Science and Engineering

Bathu, Una, HP

Bachelor of Technology in Computer Science and Engineering, **7.3 CGPA**

May 2019

INTERESTS

I like Writing, Sketching, Googling, Music, Trekking, Reading.